Annie Sun

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Education

University of British Columbia **September 2025 – April 2029**
Bachelor of Applied Science – Mechanical Engineering

Technical Skills

Mechanical Design: SolidWorks, Onshape, Fusion 360
Electronics: KiCad, Altium Designer, STM32CubeIDE
Programming & Simulation: MATLAB, Python, C, C++, ROS2

Personal Projects

Quadruped Robot **June 2026 – Present**

- Designed a compact **45:1 planetary-cycloidal gearbox** in SolidWorks for a 6 kg quadruped robot, balancing torque requirements, transmission ratios, and actuator packaging constraints.
- Developed **forward and inverse kinematics models in MATLAB** to evaluate leg workspace, joint motion, and torque requirements for locomotion.
- Designed a custom **6S BLDC motor controller PCB** in KiCad, integrating an STM32 microcontroller, three-phase MOSFET bridge, gate driver, and current sensing; ordered the initial prototype for testing.

Technical Experience

UBC Rocket **September 2026 – Present**
Avionics Hardware Member

- Developing a custom **2-layer, 915 MHz telemetry ground station PCB** using an SX1262 RF transceiver and STM32G0 microcontroller.
- Designing supporting circuitry for **USB-C power input, USB-to-UART communication, 3.3 V regulation**, and an SMA antenna interface.

Prime Robotics **July 2026 – Present**
Undergraduate Research Student

- Designed a **three-finger robotic gripper** featuring 3D-printed PLA compliant joints to accommodate variations in object geometry.
- Developed and prototyped custom PCBs integrating **inertial measurement units (IMUs)** with I2C communication for motion sensing.

UBC Uncrewed Aircraft Systems
Electrical Member *September 2026 – Present*

- Developing a custom **2.4 GHz ExpressLRS receiver**, targeting a compact 5 V design with adjustable RF transmit power.
- Researching **RF transceivers, antenna interfaces, and power management** to define system architecture and select components for a true-diversity design.

Payload Member *September 2025 – August 2026*

- Designed a **4L PETG water-carrying container**, applying thermoforming techniques and evaluating manufacturability.
- Designed a **quarter-turn locking mechanism** for rapid, secure attachment to the UAV payload frame and simplified assembly.
- Iteratively refined sealing methods, thread geometry, and mechanical tolerances to reduce leakage under dynamic flight conditions.